

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – 10

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Class: TYIT	Batch: 1
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Aim:-

a. Create a PWA With a Web App manifest file Including.

Code:-

manifest.json

```
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "447x447",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "447x447",
      "type": "image/png"
    }
  ]
}
```

Code:-

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>My PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color"
content="#0d6efd">
</head>
<body>
```

```
<h1>My Progressive Web App</h1>
<button onclick="showNotification()">
  Show Notification
</button>
<script src="app.js"></script>
</body>
</html>
```

Code:-

app.js

```
if ('serviceWorker' in navigator)
{
  navigator.serviceWorker.register('service-
worker.js')
  .then(() => console.log('Service Worker
Registered'));
}
function showNotification() {
}
Notification.requestPermission().then(permis
sion =>
{
  if (permission === 'granted')
  {
    navigator.serviceWorker.ready.then(registra
tion =>
{
  registration.showNotification('PWA
Notification', {
```

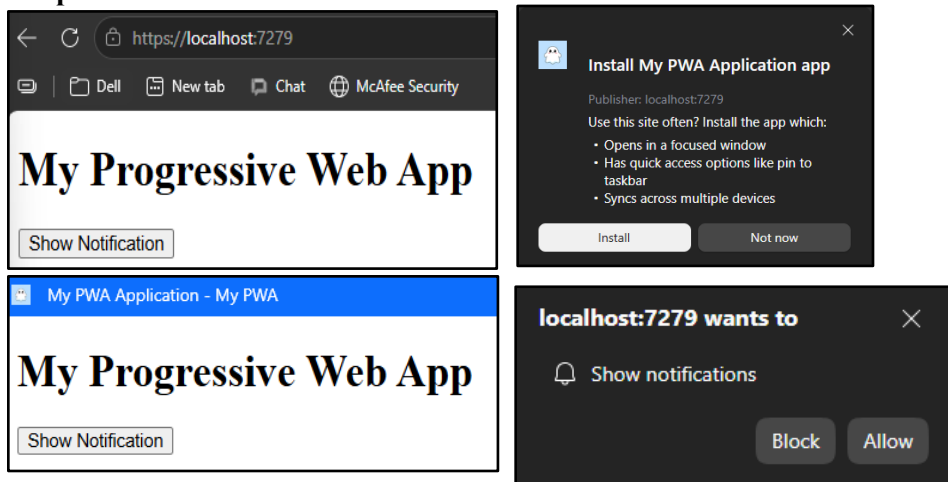
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```
        body: 'Push Notification Working  
Successfully!',  
        icon: 'icons/icon1.png'  
    });  
});  
}  
});  
Code(service-worker.js):-  
self.addEventListener('install', event =>  
{  
    console.log('Service Worker Installed');  
});
```

```
self.addEventListener('activate', event =>  
{  
    console.log('Service Worker Activated');  
});
```

Code:-
program.cs
var builder =
WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();

Output:-



b. Develop an E-Commerce PWA with product listing and offline cart functionality.

Code:-

Index.html

```
<!DOCTYPE html>  
<html>  
<head>  
    <title>E-Commerce PWA</title>  
    <link rel="manifest" href="manifest.json">  
    <meta name="theme-color" content="#0d6efd">  
</head>  
<body>  
    <div class="container">  
        <div class="product-section">  
            <div class="products">  
                <div class="product">  
                    <div class="button">  
                        <button>  
                            Add to Cart  
                        </button>  
                    </div>  
                </div>  
            </div>  
        </div>  
    </div>  
</body>  
</html>
```

```
        min-height: 300px;  
    }  
    .product-section {  
        width: 70%;  
    }  
    .products {  
        display: flex;  
        flex-wrap: wrap;  
        gap: 15px;  
    }  
    .product {  
        border: 1px solid gray;  
        padding: 10px;  
        width: 200px;  
    }  
    .button {  
        padding: 5px 10px;  
        cursor: pointer;  
    }  
</style>  
</head>
```

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```
<body>
  <h1>E-Commerce PWA</h1>
  <div class="container">
    <!-- Cart Section -->
    <div class="cart-section">
      <h2>Cart Items</h2>
      <ul id="cartList"></ul>
    </div>
    <!-- Product Section -->
    <div class="product-section">
      <h2>Products</h2>
      <div class="products">
        <div class="product">
          <h3>Laptop</h3>
          <p>Price: ₹ 50000</p>
          <button onclick="addToCart('Laptop')">Add
to Cart</button>
        </div>
        <div class="product">
          <h3>Mobile</h3>
          <p>Price: ₹ 20000</p>
          <button onclick="addToCart('Mobile')">Add
to Cart</button>
        </div>
        <div class="product">
          <h3>Headphones</h3>
          <p>Price: 3000</p>
          <button
onclick="addToCart('Headphones')">Add to Cart</button>
        </div>
      </div>
    </div>
  </div>
  <script src="app.js"></script>
</body>
</html>
```

Program.cs:-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
Manifest.json
```

```
{
  "name": "E-Commerce PWA",
  "short_name": "ShopPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
```

```
    "src": "icons/icon1.png",
    "sizes": "192x192",
    "type": "image/png"
  ],
  {
    "src": "icons/icon2.png",
    "sizes": "512x512",
    "type": "image/png"
  }
]
}
app.js
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}
let cart = JSON.parse(localStorage.getItem('cart')) || [];
displayCart();
function addToCart(product) {
  cart.push(product);
  localStorage.setItem('cart', JSON.stringify(cart));
  displayCart();
}
function displayCart() {
  let cartList = document.getElementById('cartList');
  cartList.innerHTML = "";
  cart.forEach(item => {
    let li = document.createElement('li');
    li.textContent = item;
    cartList.appendChild(li);
  });
}
```

Output:-

